Project Management

Assignment 1 - Part 2: Effort Estimation

Objective:
The goal of this assignment is to develop, compare, and critically evaluate software effort estimation models using real-world datasets. Students will apply statistical and machine learning techniques, with proper validation strategies, to understand model performance when predicting software development effort.

Task Description

- Select at least two publicly available software effort estimation datasets.
- Develop at least two estimation models, one of which must be regression-based.
- Train, validate, and compare the models using appropriate validation methodologies.
- Evaluate the performance using modern, recommended metrics (e.g., MAE, RMSE, MdAE, MASE, MdASE).
- Interpret and discuss the results.

Validation Requirements

- Use a rigorous validation strategy to avoid overfitting and ensure reliable model comparison.
- You may apply simple train–test splits (e.g., 70/30 or 80/20), but cross-validation is strongly encouraged.
- For regression-based models, employ k-fold cross-validation (e.g., k=5 or k=10) to assess generalization.
- For small datasets, prefer repeated k-fold cross-validation to reduce variance.
- Document clearly how the data was split and justify your choices.

1. Dataset Selection

- Choose at least two datasets from credible sources.
- Describe variables, size, and limitations.
- Discuss dataset characteristics that may influence model behaviour.

2. Model Development

- Describe modelling techniques and assumptions.
- Document preprocessing steps (normalization, outlier handling, etc.).
- Discuss feature selection or dimensionality reduction methods, if used.

3. Model Evaluation

- Use evaluation metrics such as MAE, RMSE, MdAE, MASE, and MdASE.
- Include tables or plots comparing model performance across datasets and validation folds.
- Discuss model stability across validation splits.

4. Analysis and Discussion

- Interpret why certain models performed better or worse.
- Discuss dataset characteristics and their impact.
- Address challenges such as skewness, variance, or multicollinearity.
- Explain implications for software project management and estimation practice.

5. Deliverables

- Submit a 5–8 page technical report including introduction, dataset description, methodology, validation strategy, results, discussion, conclusion, and appendix.
- Include code snippets only if relevant; full code may be provided as a supplementary file.

Assessment Criteria

- Correctness and rigor of model implementation.
- Quality of validation methodology and reproducibility.
- Quality of analysis and evaluation.
- Clarity of explanation and justification.
- Depth of critical reflection.
- Presentation and structure.